



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE (24AD3107) A.Y. 2026 – 2027

TITLE OF THE PROJECT	"RetailIQ" – Full-Cycle E-Commerce Customer & Sales Analytics	
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1. Introduction

E-commerce platforms generate massive volumes of transactional data every day — order details, customer demographics, product categories, pricing, discounts, and payment information. However, most businesses fail to convert this raw data into actionable insights due to a lack of structured analytical processes. **RetailIQ** is a full-cycle data analytics project that leverages a 2024–2025 e-commerce dataset to uncover patterns in customer behavior, sales performance, and product trends. By combining exploratory data analysis (EDA), customer segmentation, and predictive modeling, the project aims to demonstrate how data science techniques can turn transactional records into strategic business intelligence.

2. Problem Statement

E-commerce businesses often struggle with:

- Identifying which customers are most valuable and which are at risk of churn.
- Understanding which products or categories drive the most revenue and profit versus which underperform.
- Recognizing purchasing patterns (e.g., which products are frequently bought together) to improve cross-selling.
- Making data-driven forecasts about future sales instead of relying on intuition.

Without a systematic analytical pipeline, businesses miss opportunities for targeted marketing, inventory optimization, and revenue growth — leading to inefficient decision-making.

3. Proposed Solution

RetailIQ addresses these gaps through a structured, end-to-end analytics pipeline:

- **Data Cleaning & Preprocessing** – Handling missing values, duplicates, and inconsistent formatting to ensure a reliable dataset.
- **Exploratory Data Analysis (EDA)** – Uncovering trends in sales, revenue, discounts, and regional performance.
- **RFM Analysis (Recency, Frequency, Monetary)** – Segmenting customers into groups such as loyal, at-risk, and new customers for targeted engagement strategies.
- **Product & Sales Performance Analysis** – Ranking products/categories by revenue, profit margin, and growth trends.
- **Market Basket Analysis** – Using association rule mining (e.g., Apriori/FP-Growth) to identify frequently co-purchased products.
- **Predictive Modeling** – Applying machine learning (e.g., regression or time-series forecasting) to predict future sales trends.
- **Interactive Dashboard** – Consolidating all insights into a single visual interface with KPIs, charts, and business recommendations.

4. Abstract

RetailIQ is a comprehensive e-commerce analytics project that transforms raw transactional data into strategic business insights. Using a 2024–2025 dataset spanning orders, customers, products, and sales, the project applies data cleaning, exploratory analysis, RFM-based customer segmentation, market basket analysis, and machine learning-based sales prediction. The processed insights are visualized through an interactive dashboard, enabling stakeholders to monitor key performance indicators, identify high-value customer segments, track product trends, and receive data-backed recommendations. This project demonstrates the practical application of the data science lifecycle — from raw data to decision-ready insights — in a real-world retail context.

5. Tools & Technologies Used

Category	Tools
Programming Language	Python
Data Manipulation	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Machine Learning	Scikit-learn
Market Basket Analysis	mlxtend (Apriori/FP-Growth)
Dashboard	Power BI / Streamlit (choose based on your submission requirement)
Database (optional)	SQL/MySQL for data storage and querying
Environment	Jupyter Notebook

6. Conclusion

RetailIQ demonstrates how a structured data science approach can convert unorganized e-commerce data into meaningful, decision-ready insights. By integrating customer segmentation, sales analysis, market basket analysis, and predictive modeling into a single interactive dashboard, the project provides a holistic view of business performance. This solution can help e-commerce businesses improve customer retention, optimize product offerings, and make informed, data-driven decisions — ultimately supporting sustainable business growth.

Signature of the Guide

HOD